Tower F4, signature F5 and an F6 analogue on prime-field curves

Run report, ecbench, 2026-10-11 22:44 UTC onward (run run-20261011T2244Z)

1. What ran

- **Protocol.** research/pkm_tower_ecbench_20261011/PROTOCOL.md, committed before any measured run, and AMENDMENT_1.md.
- **First attempt.** It started at 22:44 UTC from commit 280c5ad52. Its P6 replay audit failed because the tower F4 count depended on the thread count. It is kept in attempt-1-voided/, and none of its numbers enters this report.
- **Measured attempt.** It started at 22:51 UTC from commit faeeae26a. Every session records that commit, a clean tree, and ecbench binary sha256 64ed18283b4e8215….
- **Host.** An x86-64 Linux cloud VM. The isolation earned per run, L1 or L2, is recorded per arm in ANALYSIS.json.
- **Arms.** All IC arms use ic.pipeline with prime-tower:t=T,seed=1 and pkm-tower:m=2,t=T,seed=1,engine=E,degree=20, max_trials=200000:
  - f4: tower F4 decides each node of the splitting tree.
  - sig-f5: signature F4 with the F5/GVW criteria decides each node.
  - f6: tower F4, with nodes of at most 4 candidates closed by charged group additions.
  - enum: closure at the root; no algebra.

  Reference: rho.negation on the same target.
- **Seeds and workloads.** Target seed 202610112301, measurement seed 202610112302, one round. Public hash_to_subgroup_v1 targets: 8 per curve, 4 at P8.

spec	session	curve	p	r	t	verified	replay
P5	ECBS1h744c8f794e9a	icv1-fp9-t25-33fa1ca2	257	233	5	40/40	ok
P6	ECBS1h91a14b12224a	icv1-fp11-t43-4d73e7f7	1,409	1,367	6	40/40	ok
P7	ECBS1h4e2538d423f5	icv1-fp13-tm41-756f569a	7,681	7,723	7	40/40	ok
P8	ECBS1ha6de54733ae8	icv1-fp15-t181-50ff72d2	18,433	18,253	8	20/20	ok

Raw records: sessions/<spec>/records.jsonl. Replay receipts: audits/<spec>-replay-all.json. Derived values: ANALYSIS.json, written by examples/pkm_tower_prime_analyze.rs from the records alone.

2. Measured values

Units.

- W: counted F_p multiply-adds per run (measured, deterministic).
- S: charged GAE over √r (measured). It is a lower bound for the tower arms, whose multiply-adds are unpriced.
- Medians are derived by the analyser.

cell	rho S	F4 S	sig-F5 S	F6 S	enum S	F4 W	sig-F5 W	F6 W
P5	5.50	25.84	25.84	25.55	29.74	8.05e7	1.82e7	8.02e7
P6	2.79	15.55	15.55	15.67	38.81	2.35e9	6.51e8	2.35e9
P7	2.01	9.37	9.37	9.38	45.89	4.07e10	1.25e10	4.07e10
P8	1.98	7.48	7.48	7.50	60.62	4.86e11	1.59e11	4.86e11

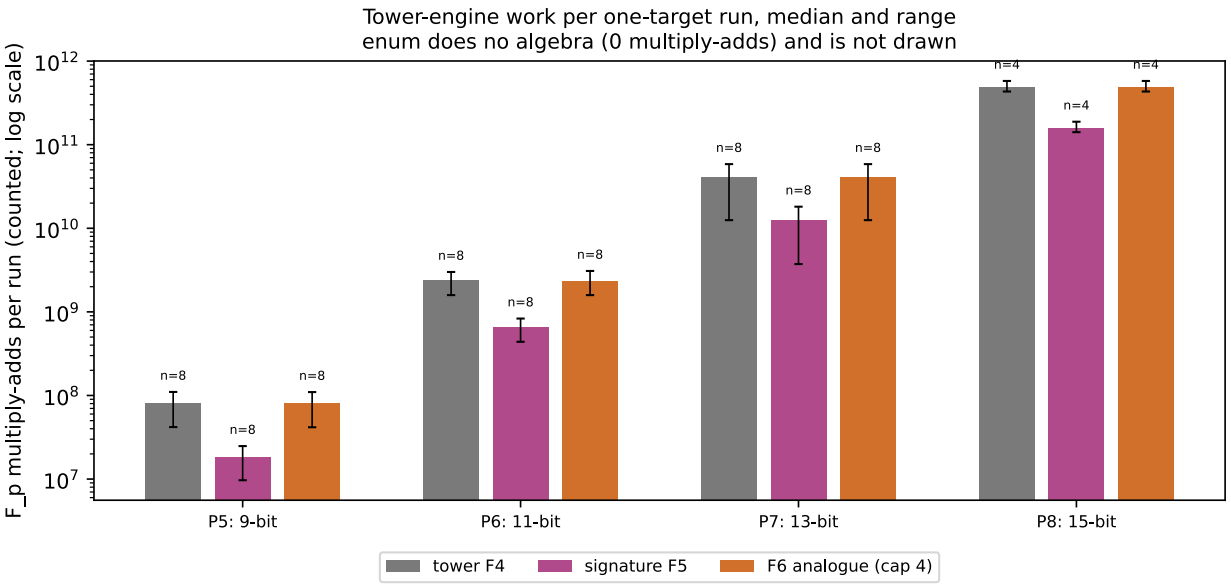


Figure 1: Counted `F_p` multiply-adds per run for the three tower engines. Bars show the median and ticks the range over 8 targets (4 at P8), for sessions P5–P8.

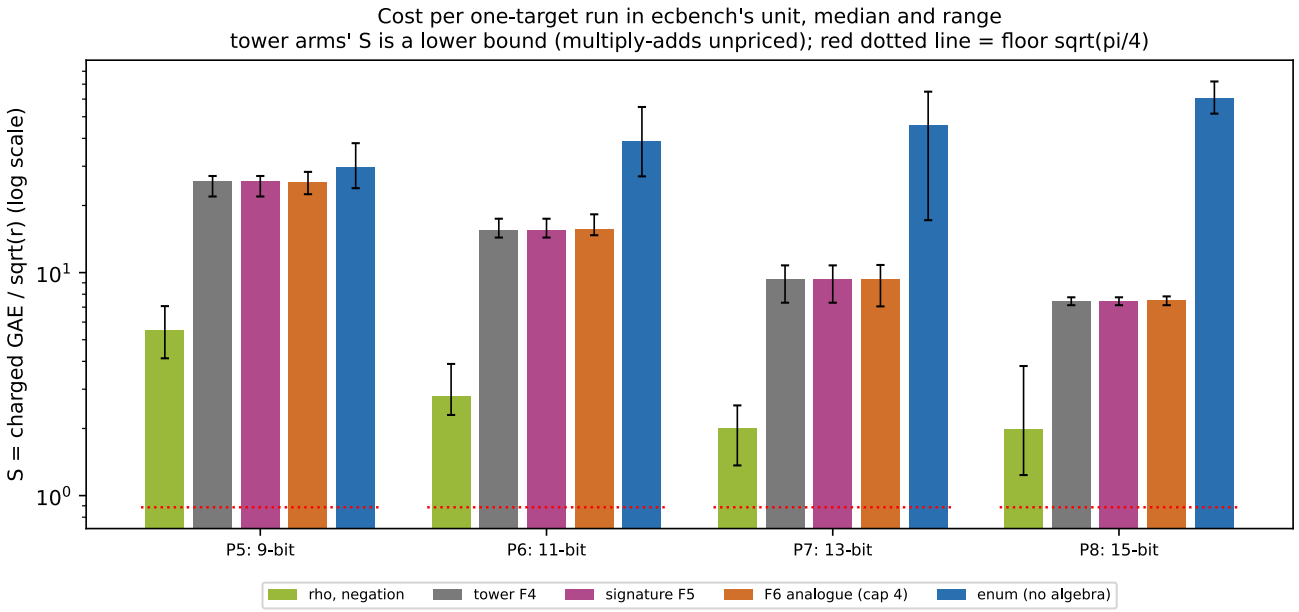


Figure 2: Lower-bound `S` per arm and cell, with the median and range. The red dotted line is the floor $\sqrt{(\pi/4)}$. The tower arms' `S` omits their multiply-adds.

3. The protocol's questions (derived; rules applied as written)

- **Q1. Signature F5 / F4 multiply-adds.**
  - Medians: 0.227, 0.277, 0.307, 0.326 (P5 ⟶ P8).
  - The tree is the same on every target.
  - Rule: **F5 cuts F4's work** in every cell.
  - Relation wall, signature F5 / F4: 0.56, 0.94, 1.45, 2.06.
- **Q2. $\log_2$(W F4 / W F6).**
  - Medians: 0.0059, 0.0005, 0.00007, 0.00001.
  - Rule: **no saving** in every cell.
  - Closure removes 27–47 % of the nodes, but almost none of the work.
- **Q3. S / same-target rho.** Every arm is above 1 on every target. Rule: **slower than rho on every target**.
  - F4: 4.64, 5.39, 4.77, 3.91.
  - `enum`: 5.42, 12.0, 21.7, 29.0.
- **Q4. Algebra / `enum`.**
  - Lower-bound `S(F4)/S(enum)`: 0.85 ⟶ 0.12.
  - Derived, with multiply-adds priced at 1/63 GAE:

  - F4: 2,949 ⟶ 899,485;
  - signature F5: 670 ⟶ 293,124.
- Relation wall, F4 / `enum`: 8,041 ⟶ 360,005.

- **Q5. Slopes of $\log_2 S$ against $\log_2 r$** (descriptive, four cells): rho −0.24, F4 −0.29, F6 −0.28, `enum` +0.15.

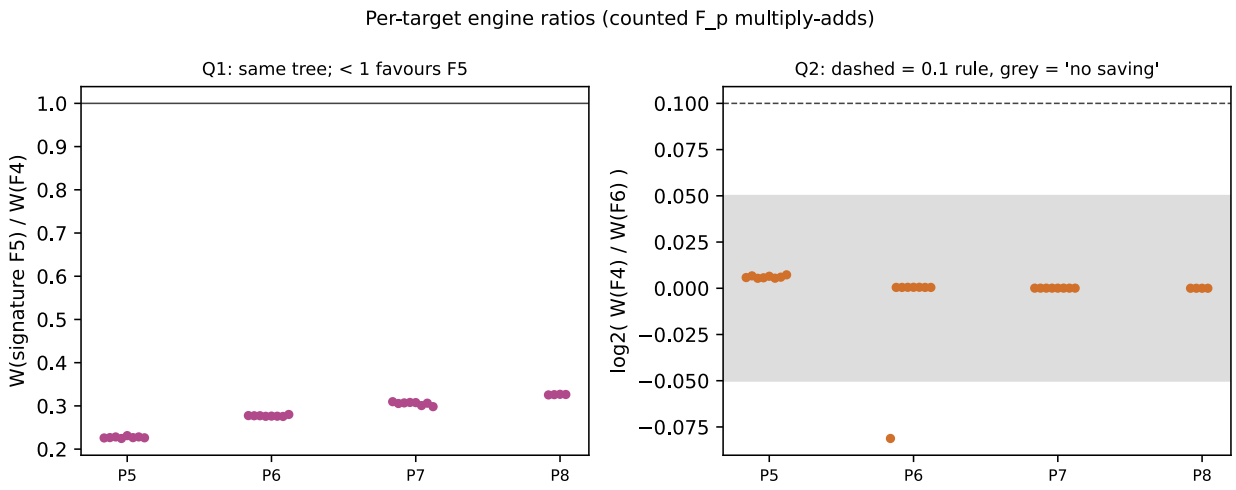


Figure 3: Per-target engine ratios in counted multiply-adds. Left: signature F5 / F4 on identical trees. Right: $\log_2$(F4 / F6), with the preregistered 0.1 threshold and the "no saving" band.

4. Limits

- **Wall time** is a practicality note from mixed L1/L2 runs.
- **Unpriced work.** The tower arms' multiply-adds are counted but not priced, and lookups and square roots are unpriced for every IC arm.
- **Scope.** `m = 2`, Kummer towers, one splitting rule; four sizes up to 15 bits.
- **No claims.** No exponent, speedup or cryptographic-size claim is made. Interpretation is in the study's `RESULT.md`.